

5. Macro-Factor Investing

So far we have taken an investment, or style, approach to factor investing as our factors have all been tradeable investing strategies. An alternative approach to factor investing can be based on (nontraded) macro factors. This is academically purer but much harder to implement.

This is because asset classes do not move one-for-one with macro factors, and in fact, many of their movements are perverse or at least unintuitive. Equities, for example, are a claim on real assets yet a terribly inadequate choice for tracking inflation (see chapter 8). Real estate is a better inflation hedge but it is only partially “real” (see chapter 11). Bond prices reflect inflation risk, but other factors, including monetary policy risk and illiquidity risk are important drivers of long-term bonds (see chapter 9).

Academic Model

Taking a macro view requires a framework for how macro factors simultaneously affect many asset classes. Ang and Ulrich (2012) is one model that could be used for macro-factor investing. We develop a model where real bonds, nominal bonds, and equities are jointly determined by two macro factors—inflation and economic growth—along with a factor capturing how the Fed sets monetary policy. If there is an inflation shock, for example, the model shows how the prices of all asset prices move. Macro factors are responsible for a lot of the variation in expected returns; for example, economic growth accounts for approximately 60% of variation in the equity risk premium, while expected inflation accounts for 40% of variation in real rates and 90% of variation in nominal bonds.

Bridgewater Associates

A practitioner framework where macro factors affect asset prices has been developed by Bridgewater Associates, a large hedge fund. Bridgewater’s All Weather Strategy considers inflation and economic growth as macro factors, as shown in Figure 14.8. Although Bridgewater deviated from its originally intended strategy during the financial crisis, it is worth examining their overall framework. Bridgewater estimates how different assets perform during periods when growth is high or low, or when inflation is rising or falling. They fill in a 2×2 matrix corresponding to these scenarios. In Figure 14.8, they expect that nominal bonds do well when growth and inflation are falling. Equities, on the other hand, outperform when growth is high and inflation is subdued. This is clearly a much more simplified version compared to a formal model like Ang and Ulrich (2012), but the idea is the same: we want to see how assets react when macro factors change.